Anusha Devulapally

+1 (814) 826 8627 | akd5994@psu.edu | Linkedin:anusha-devulapally | Google Scholar | GitHub

Education

The Pennsylvania State University — University Park, PA
Ph.D. in Computer Science and Engineering — Aug 2021 – Jan 2027 (Expected)
Advisor: Dr. Vijaykrishnan Narayanan
Dissertation: *Multi-Modal AI Models and Memory-Centric Analysis for Emerging Architectures*

Indian Institute of Technology, Goa — Goa, India
M.Tech in Computer Science and Engineering — Jul 2019 – Aug 2021
Advisors: Dr. Clint P. George, Dr. Sreenath Balakrishnan — **CGPA:** 8.42/10.00
Thesis: *Cell Segmentation from 3D Confocal Images using U-Net Variants*

Experience

AI/HPC System Research Intern — Aug 2026 – Nov 2026
SK Hynix America

- Developed a GPU–CXL–NVMe memory hierarchy for large-scale GNN training, enabling direct GPU access to CXL-resident features while using SSD as the backing store for datasets exceeding GPU HBM capacity.
- Designed hot-feature placement, asynchronous prefetching, and cache/victim management across GPU HBM and CXL memory to reduce demand SSD accesses and feature-fetch latency.

Graduate Research Assistant — Aug 2021 – Present
Microsystems Design Lab, Penn State — *University Park, PA*

- Developed adaptive multi-modal models for scientific perception tasks, with emphasis on robustness under missing or noisy inputs, efficient model execution, and rigorous evaluation.
- Built transformer-based models for multimodal monocular depth estimation using event and RGB inputs, achieving state-of-the-art performance with an 8% reduction in mean depth error.
- Profiled and analyzed large-scale AI workloads on CXL-based disaggregated memory systems, identifying latency, bandwidth, and scalability bottlenecks for inference workflows, including GPU-accelerated and multi-node settings.

Ph.D. Research Intern — May 2024 – Nov 2025
Pacific Northwest National Laboratory — *Richland, WA*

- Built cross-modal validation and refinement pipelines for scientific document understanding and segmentation on geo-referenced NEPA datasets.
- Fine-tuned OpenCLIP for caption generation and image-to-text retrieval on large scientific document collections related to wind energy.
- Designed and evaluated analytical and learning-based scheduling frameworks for disaggregated memory systems, improving I/O efficiency for AI workloads.
- Implemented and benchmarked greedy and reinforcement learning-based schedulers for large computational workflows, with a focus on algorithmic efficiency and system-level evaluation.

Technical Skills

Programming: Python, C/C++, MATLAB
Machine Learning & AI: PyTorch, Transformers, Vision-Language Models (VLMs), Multimodal Learning, OpenCLIP, Spiking Neural Networks (SNNs)
Scientific Computing & Data: NumPy, pandas, scikit-learn, large-scale dataset handling, data preprocessing, statistical data analysis, benchmarking
Systems & HPC: Disaggregated Memory Systems (CXL), Computer Architecture, Performance Modeling, HPC Workloads, OpenMP, CUDA, multi-node training, workflow scripting
Software & Tools: Linux, Git, Jupyter, OpenCV, TensorFlow
Optimization: Profiling, Quantization, Compression, Latency-Accuracy Trade-offs, RL-based optimization

Publications

Systems & Hardware Publications

- [**MEMSYS 2025**] **Anusha Devulapally**, Mahantesh Halappanavar, Bruno Jacob, Vijaykrishnan Narayanan, Andres Marquez, **"Revisiting Pebble Games for Modeling and Efficient Use of Disaggregated Memory Systems."**
- [**ISPASS 2025**] Pingyi Huo, **Anusha Devulapally**, Hasan Al Maruf, Meena Arunachalam, Mahmut Taylan Kandemir, Vijaykrishnan Narayanan, **"TPNM: A CXL Based General Purpose Tiered Process Near Memory Framework."**
- [**MICRO 2025**] Pingyi Huo, **Anusha Devulapally**, Hasan Al Maruf, Nandhini Chandramoorthy, Meena Arunachalam, Gulsum Gudukbay Akbulut, Mahmut Taylan Kandemir, Vijaykrishnan Narayanan, **"Multi-Dimensional ML-Pipeline Optimization in Cost-Effective Disaggregated Datacenter."**
- [**MEMSYS 2024**] **Anusha Devulapally**, Mahantesh Halappanavar, Amit Puri, Vijaykrishnan Narayanan, Andres Marquez, **"Using Isoefficiency as a Metric to Assess Disaggregated Memory Systems for High Performance Computing."**
- [**MICRO 2024**] Pingyi Huo, **Anusha Devulapally**, Hasan Al Maruf, Minseo Park, Krishnakumar Nair, Meena Arunachalam, Gulsum Gudukbay Akbulut, Mahmut Taylan Kandemir, Vijaykrishnan Narayanan, **"PIFS-Rec: Process-In-Fabric-Switch for Large-Scale Recommendation System Inferences."**

Vision & VLM Publications

- [**AI4ChipDesign 2026 -Neurips**] Miyuki Tokuhara, **Anusha Devulapally**, Varun Darshana Parekh, Vijaykrishnan Narayanan, **"PhysThermix: A Physics-Informed Surrogate for Thermal Prediction of 2.5D Chiplet Architectures"**
- [**NCE-IOP Science**] **Anusha Devulapally**, Sadia Anjum Tumpa, Vijaykrishnan Narayanan, **"A Review of Event-Based Vision Sensor Fusion: Architectures, Evaluation and Robustness Challenges"**
- [**CVPRW 2024**] **Anusha Devulapally**, Md Fahim Faysal Khan, Siddharth Advani, Vijaykrishnan Narayanan, **"Multi-Modal Fusion of Event and RGB for Monocular Depth Estimation Using a Unified Transformer-based Architecture."**
- [**ACM MM 2022**] Md Fahim Faysal Khan, **Anusha Devulapally**, Siddharth Advani, Vijaykrishnan Narayanan, **"Robust multimodal depth estimation using transformer based generative adversarial networks."**
- [**AGU 2025**] **Anusha Devulapally**, Heng Wan, Cleve Davis, Benjamin Chauhan, Micah Taylor, Anurag Acharya, Anastasia Bernat, Sai Munikoti, Sameera Horawalavithana, **"Cross-Modal Validation and Refinement for Scalable Digitization of NEPA Map Features."**
- [**AGU 2025**] Heng Wan, **Anusha Devulapally**, Cleve Davis, Sridevi Wagle, Benjamin Chauhan, Micah Taylor, Anurag Acharya, Anastasia Bernat, Sai Munikoti, Sameera Horawalavithana, **"Building the Foundation for Georeferencing: A Preprocessing Workflow for NEPA Map Images."**
- [**ICONS 2025**] Sadia Anjum Tumpa, **Anusha Devulapally**, Chonghan Lee, Chita R. Das, Vijaykrishnan Narayanan, **"DESTformer: Energy-Efficient Monocular Depth Estimation with Spiking Transformer for Edge Devices."**
- [**VLSI Design 2025**] **Anusha Devulapally***, Varun Parekh*, Sivani Devarapalli, Cassius Henderson, Vijaykrishnan Narayanan, **"ThermLeT: Transformer based Temperature Prediction for 2.5D Chiplet Architecture."** (*equal contribution)
- [**ISVLSI 2024**] Sadia Anjum Tumpa, **Anusha Devulapally**, Matthew Brehove, Espoir Kyubwa, Vijaykrishnan Narayanan, **"SNN–ANN Hybrid Networks for Embedded Multimodal Monocular Depth Estimation."**
- [**Cells Tissues Organs (CTO) 2024**]**Anusha Devulapally**, Varun Darshana Parekh, Clint P. George, Sreenath Balakrishnan, **"On the Variability in Cell and Nucleus Shape."**

Manuscripts in Submission

- [**CVPR 2026**] **Anusha Devulapally**, Vijaykrishnan Narayanan, **"Conditional RGB–Event Fusion for Monocular Depth Estimation with Foundation Models"**
- [**DAC 2026**] Varun Darshana Parekh, **Anusha Devulapally**, Vijaykrishnan Narayanan, **"Accelerating Speculative Decoding on 2.5D Heterogeneous Integration Systems"**
- [**iTherm 2027**] Varun Darshana Parekh, Miyuki Tokuhara,**Anusha Devulapally**, Srivatsa Rangachar Srinivasa, Mirco Sciulli, Vijaykrishnan Narayanan, **"Physics-Informed Transformer Surrogate for Thermal Prediction across 2.5D Chiplet Architectures"**

Talks

- **Oral presentations:** MULA Workshop at CVPR 2024 (Seattle, WA); MEMSYS 2024 and MEMSYS 2025 (Washington, D.C.).
- **Invited talks:** SRC TECHCON (2023); PRISM Liaison Meeting (2025); CBRIC Industry Meeting (2022).
- **Poster:** SRC PRISM Annual Review, UC San Diego, CA (2024).

Mentorship

- **Undergraduate students:** mentored two SROP (Summer Research Opportunities Program) students in 2025 and 2026 on **ThermLeT** and its physics-informed variant, each contributing to a publication (VLSI Design 2025; NeurIPS AI for Chip Design Workshop 2026); guided an undergraduate capstone on **mapping AI workloads to edge devices**; advised two undergraduates at **SRC TECHCON 2025**.
- **Graduate student:** advised a master's student on **AI/optimization-driven chip placement**.
- **High-school teacher:** mentored a teacher through Penn State's RET program (2026) on AI-assisted native plant recommendations for pollinator support.

Awards and Service

- **Award:** Honorable Award, AI Week Poster Session, AI Hub, Penn State (2024).
- **Peer Reviewer:** ICRA, IEEE TVLSI, IEEE TPDS, NeurIPS, IEEE TMM, ACM TOMM, IEEE TAI, CVIU, Neurocomputing, Information Processing in Agriculture, ICISN.
- **Judge:** Penn State Undergraduate Exhibition (2025). **Member:** IEEE (Student Member).
- **Volunteer:** ASPLOS 2026; Girls Who Code (2023); Treasurer, IEEE Student Body, IIT Goa; First Virtual Research Scholar's Confluence, IIT Goa.